

Er. Abhishek Halwai

Environmental Engineer

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Environmental Engineer specializing in anaerobic digestion and engineered bioprocess systems for resource recovery. Experienced in pilot-scale reactor operation, biomethane potential (BMP) assays, trace element optimization, and quantitative analysis of methane production kinetics. Interested in linking microbial community dynamics with reactor performance through systems-level process engineering and modeling approaches.

EXPERIENCE

Waste Smart Fellow

Youth Innovation Lab

Janakpurdham Sub Metropolitan City

Jun, 2025 – Sep, 2025

Collaborated with Janakpur Sub-Metropolitan City to advance sustainable waste management and circular economy practices through policy, community engagement, and localized solutions.

- Coordinated with municipal officers, ward representatives, and sanitation staff to strengthen plastic and solid waste management practices.
- Organized stakeholder sensitization sessions on circular economy, source segregation, and climate change.

Research Assistant

Resource Recovery Research Laboratory

Kathmandu University, Dhulikhel

Jun, 2023 – Sep, 2024

Project Title: "Optimization of Resources Recovery from Waste Biomass via Dark, Photo-fermentation, and Anaerobic Digestion: An Integrated Circular Biorefinery (ORWIC)."

Key Responsibilities:

- Operated and monitored pilot-scale mesophilic anaerobic digestion reactors treating food waste.
- Conducted triplicate biomethane potential (BMP) assays and analyzed cumulative methane production kinetics.
- Evaluated effects of trace element supplementation on methanogenic activity and reactor stability.
- Performed chemical analyses including COD, VFA, alkalinity, and nutrient profiling.
- Interpreted process stability using VFA/Alkalinity ratio and gas composition trends.
- Contributed to manuscript preparation for peer-reviewed journals.

EDUCATION

B.Tech in Environmental Engineering

Kathmandu University

Dhulikhel

Aug, 2018 – May, 2023

CGPA: 3.58/4.00

Thesis: Effect of Trace Element Supplementation on the Mesophilic Anaerobic Digestion of Food Waste

Intermediate in Science

Holyland Higher Secondary School

Nepalgunj

Nov, 2015 - Nov, 2017

PROJECTS

Optimization of Resources Recovery from Waste Biomass via Dark, Photo Fermentation and Anaerobic Digestion (ORWIC)

Effect of Trace Element Supplementation on Methanogenic Activity and Process Stability in Mesophilic Anaerobic Digestion of Food Waste

ACHIEVEMENTS

Research Output and Presentations

- Research Paper (Under Review): Effect of Trace Element Supplementation on Mesophilic Anaerobic Digestion of Food Waste.
- Conference Presentation: Poster presented at IWA Conference, Turkey, on trace element-enhanced biogas production and reactor performance stability.

Training and Certifications

Environmental Engineer

2023

Nepal Engineering Council

NEC Reg No. 79207

Excel Essential Training

LinkedIn

SKILLS

- Anaerobic Digestion & Bioreactor Operation
- Biomethane Potential (BMP) Assays
- Process Stability Analysis (VFA/Alkalinity, pH, Gas Composition)
- Trace Element Optimization in AD Systems
- Chemical Analysis (COD, TS/VS, Nutrient Profiling)
- Data Analysis & Visualization (Python, Excel)
- Regression & Trend Analysis of Methane Production Curves

LANGUAGES

English, Nepali, Hindi, Swedish, German